

Working Document: Complete Source Reference List

Literature Review on Vision-Language Models for Construction Site Monitoring

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Submitted as required element of Assignment 11 (Final Literature Review Manuscript); supersedes the Assignment 10 version

Corpus Overview

The total corpus for this scoping review comprises **58 peer-reviewed studies**, reconciled from the authoritative charted dataset and assembled via two parallel identification tracks. The Phase 1 database track (Web of Science and Scopus, searched 4 March 2026, followed by title/abstract and full-text screening) yielded **19 core VLM studies**, classified into Paradigms A, B, and C. The Phase 2 backward-snowballing track yielded **39 background/transition studies**, of which 5 are transformer-based transitional “bridging” works and 34 are pre-VLM baseline studies (sensor-based, computer vision, or NLP approaches).

Of the 34 Phase 2 baseline studies, 17 are cited individually in the manuscript; the remaining 17 appear only in aggregate as contextual background. All 42 sources individually cited in the manuscript are listed below in full bibliographic detail, organized by corpus phase and architectural paradigm. The additional 17 Phase 2 baseline papers are tracked in the Phase 2 evidence synthesis matrix from Assignment 8.

Note on the arXiv preprint by Bui et al. (2026): this source is *not* counted in the 58-study corpus and contributes no descriptive or quantitative figures, consistent with the manuscript’s peer-reviewed-only eligibility criterion. It is retained as a single, clearly flagged piece of contextual evidence in the Discussion (its pre-review status is noted wherever cited) and is listed separately below rather than under Paradigm A.

1 Phase 1 Core VLM Studies (19 papers)

1.1 Paradigm A: Zero- and Few-Shot Retrieval (6 papers)

- A1.** Zhang, Z., Yu, Y., Pan, Z., and Antwi-Afari, M. F. (2025). “Training-free few-shot construction tool and material detection using pre-trained vision-language model.” *Computer-Aided Civil and Infrastructure Engineering*, 40(30), 6004–6023. DOI: 10.1111/mice.13326
- A2.** Gil, M., and Lee, S. (2024). “Zero-shot monitoring of construction workers’ personal protective equipment based on image captioning.” *Automation in Construction*, 164, 105470. DOI: 10.1016/j.autcon.2024.105470
- A3.** Liang, Y., et al. (2024). “Recognizing temporary construction site objects using CLIP-based

few-shot learning and multi-modal prototypes.” *Automation in Construction*, 165, 105542. DOI: 10.1016/j.autcon.2024.105542

- A4.** Wang, X., and El-Gohary, N. (2024). “Few-shot object detection and attribute recognition from construction site images for improved field compliance.” *Automation in Construction*, 167, 105539. DOI: 10.1016/j.autcon.2024.105539
- A5.** Chen, C., Xiao, B., Zhang, Y., and Zhu, Z. (2023). “Automatic vision-based calculation of excavator earthmoving productivity using zero-shot learning activity recognition.” *Automation in Construction*, 146, 104702. DOI: 10.1016/j.autcon.2022.104702
- A6.** Sun, Y., Gu, Z., and Yang, S. B. (2024). “Probing vision and language models for construction waste material recognition.” *Automation in Construction*, 166, 105629. DOI: 10.1016/j.autcon.2024.105629

Contextual Evidence — Not Counted in Corpus

- X1.** Bui, H. T., Chodosh, N. E., and Tavakoli, A. (2026). “Can vision-language models understand construction workers? An exploratory study.” arXiv preprint arXiv:2601.10835 (not peer-reviewed at time of search; cited only as contextual evidence, see Corpus Overview). arXiv: 2601.10835

1.2 Paradigm B: Domain Fine-Tuned Generative Models (6 papers)

- B1.** Liu, H., Wang, G., Huang, T., He, P., Skitmore, M., and Luo, X. (2020). “Manifesting construction activity scenes via image captioning.” *Automation in Construction*, 119, 103334. DOI: 10.1016/j.autcon.2020.103334
- B2.** Xiao, B., Wang, Y., and Kang, S.-C. (2022). “Deep learning image captioning in construction management: A feasibility study.” *Journal of Construction Engineering and Management*, 148(7), 04022049. DOI: 10.1061/(ASCE)CO.1943-7862.0002297
- B3.** Zhai, P., Wang, J., and Zhang, L. (2023). “Extracting worker unsafe behaviors from construction images using image captioning with deep learning-based attention mechanism.” *Journal of Construction Engineering and Management*, 149(2), 04022164. DOI: 10.1061/JCEMD4.COENG-12096
- B4.** Wang, Y., Xiao, B., Bouferguene, A., and Al-Hussein, M. (2024). “Proactive safety hazard identification using visual-text semantic similarity for construction safety management.” *Automation in Construction*, 166, 105602. DOI: 10.1016/j.autcon.2024.105602
- B5.** Jung, Y., Cho, I., Hsu, S.-H., and Golparvar-Fard, M. (2024). “VisualSiteDiary: A detector-free vision-language transformer model for captioning photologs for daily construction reporting and image retrievals.” *Automation in Construction*, 165, 105483. DOI: 10.1016/j.autcon.2024.105483
- B6.** Bang, S., and Kim, H. (2020). “Context-based information generation for managing UAV-acquired data using image captioning.” *Automation in Construction*, 112, 103116. DOI: 10.1016/j.autcon.2020.103116

1.3 Paradigm C: Hybrid Geometric-Semantic / Agentic Systems (7 papers)

- C1.** Chen, H., Hou, L., Wu, S., Zhang, G., Zou, Y., Moon, S., and Bhuiyan, M. (2024). “Augmented reality, deep learning and vision-language query system for construction worker safety.” *Automation in Construction*, 157, 105158. DOI: 10.1016/j.autcon.2023.105158
- C2.** Hussain, R., Lee, D., Abbas, M. S., Zaidi, S. F. A., Pedro, A., and Park, C. (2026). “Vision-language model-based intelligent assistant for onsite construction safety inspection.” *Automation in Construction*, 182, 106728. DOI: 10.1016/j.autcon.2025.106728
- C3.** Chan, C.-F., Wong, P. K.-Y., Guo, X., Cheng, J. C. P., Chan, J. P.-C., Leung, P.-H., and Tao, X. (2025). “Context-aware vision-language model agent enriched with domain-specific ontology for construction site safety monitoring.” *Automation in Construction*, 177, 106305. DOI: 10.1016/j.autcon.2025.106305
- C4.** Xiao, B., Wang, Y., Zhang, Y., Chen, C., and Darko, A. (2024). “Automated daily report generation from construction videos using ChatGPT and computer vision.” *Automation in Construction*, 168, 105874. DOI: 10.1016/j.autcon.2024.105874
- C5.** Jeoung, H., Jung, M., and Kim, C. (2025). “Zero-shot framework for construction equipment task monitoring.” *Computer-Aided Civil and Infrastructure Engineering*. DOI: 10.1111/mice.13506
- C6.** Tuomisto, J., Zheng, Y., Chauhan, I., Seppänen, O., and Pajarinen, J. (2026). “Automating on-site object inspection with a quadruped robot and BIM.” *Automation in Construction*, 181, 106571. DOI: 10.1016/j.autcon.2025.106571
- C7.** Wang, B., Chen, Z., Li, M., Wang, Q., Yin, C., and Cheng, J. C. P. (2024). “Omni-Scan2BIM: A ready-to-use Scan2BIM approach based on vision foundation models for MEP scenes.” *Automation in Construction*, 162, 105384. DOI: 10.1016/j.autcon.2024.105384

2 Phase 2: Transitional Bridging Studies (5 papers)

Identified via backward snowballing, these transformer-based studies bridge the unimodal-to-multimodal transition and are classified under the Phase 2 (snowball) track in the reconciled PRISMA-ScR flow.

- T1.** Yang, M., Wu, C., Guo, Y., Jiang, R., Zhou, F., Zhang, J., and Yang, Z. (2023). “Transformer-based deep learning model and video dataset for unsafe action identification in construction projects.” *Automation in Construction*, 146, 104703. DOI: 10.1016/j.autcon.2022.104703
- T2.** Jang, J., Cha, M., Park, M., and Kim, H. (2025). “Transformer-based deep learning model and video dataset for installation action recognition in offsite projects.” *Automation in Construction*, 172, 106042. DOI: 10.1016/j.autcon.2025.106042
- T3.** Xin, C., Hartel, A., and Kasneci, E. (2024). “DART: An automated end-to-end object detection pipeline with data diversification, open-vocabulary bounding box annotation, pseudo-label review, and model training.” *Expert Systems with Applications*, 258, 125124. DOI: 10.1016/j.eswa.2024.125124

- T4.** Tohidifar, A., Kim, D., and Lee, S. (2024). “Make it till you fake it: Construction-centric computational framework for simultaneous image synthetization and multimodal labeling.” *Automation in Construction*, 167, 105696. DOI: 10.1016/j.autcon.2024.105696
- T5.** Xiong, R., Song, Y., Li, H., and Wang, Y. (2019). “Onsite video mining for construction hazards identification with visual relationships.” *Advanced Engineering Informatics*, 42, 100966. DOI: 10.1016/j.aei.2019.100966

3 Phase 2 Baseline Studies — Individually Cited (17 of 34 papers)

These Phase 2 studies were identified via backward snowballing to establish the pre-VLM baseline and are individually cited in the manuscript at specific analytical points. The remaining 17 of the 34 baseline studies appear only in aggregate (Phase 2 evidence synthesis matrix, Assignment 8).

- P2-1.** Alaloul, W. S., Alzubi, K. M., Malkawi, A. B., Al Salaheen, M., and Musarat, M. A. (2022). “Productivity monitoring in building construction projects: A systematic review.” *Engineering, Construction and Architectural Management*, 29(7), 2760–2785. DOI: 10.1108/ECAM-03-2021-0211
- P2-2.** Asadzadeh, A., Arashpour, M., Li, H., Ngo, T., Bab-Hadiashar, A., and Rashidi, A. (2020). “Sensor-based safety management.” *Automation in Construction*, 113, 103128. DOI: 10.1016/j.autcon.2020.103128
- P2-3.** Bohn, J. S., and Teizer, J. (2010). “Benefits and barriers of construction project monitoring using high-resolution automated cameras.” *Journal of Construction Engineering and Management*, 136(6), 632–640. DOI: 10.1061/(ASCE)CO.1943-7862.0000167
- P2-4.** Davila Delgado, J. M., and Oyedele, L. (2021). “Deep learning with small datasets: Using autoencoders to address limited datasets in construction management.” *Applied Soft Computing*, 112, 107836. DOI: 10.1016/j.asoc.2021.107836
- P2-5.** Jiang, Z., and Messner, J. I. (2023). “Computer vision applications in construction and asset management phases: A literature review.” *Journal of Information Technology in Construction (ITcon)*, 28(9), 176–199. DOI: 10.36680/j.itcon.2023.009
- P2-6.** Li, H., Deng, H., and Deng, Y. (2025). “Towards worker-centric construction scene understanding: Status quo and future directions.” *Automation in Construction*, 171, 106005. DOI: 10.1016/j.autcon.2025.106005
- P2-7.** Nath, N. D., Behzadan, A. H., and Paal, S. G. (2020). “Deep learning for site safety: Real-time detection of personal protective equipment.” *Automation in Construction*, 112, 103085. DOI: 10.1016/j.autcon.2020.103085
- P2-8.** Paneru, S., and Jeelani, I. (2021). “Computer vision applications in construction: Current state, opportunities & challenges.” *Automation in Construction*, 132, 103940. DOI: 10.1016/j.autcon.2021.103940
- P2-9.** Parasuraman, R., Sheridan, T. B., and Wickens, C. D. (2000). “A model for types and levels of

human interaction with automation.” *IEEE Transactions on Systems, Man, and Cybernetics – Part A*, 30(3), 286–297. DOI: 10.1109/3468.844354

- P2-10.** Rabbi, A. B. K., and Jeelani, I. (2024). “AI integration in construction safety: Current state, challenges, and future opportunities in text, vision, and audio based applications.” *Automation in Construction*, 164, 105443. DOI: 10.1016/j.autcon.2024.105443
- P2-11.** Samsami, R. (2024). “A systematic review of automated construction inspection and progress monitoring (ACIPM): Applications, challenges, and future directions.” *CivilEng*, 5(1), 265–287. DOI: 10.3390/civileng5010014
- P2-12.** Sherafat, B., Ahn, C. R., Akhavian, R., Behzadan, A. H., Golparvar-Fard, M., Kim, H., Lee, Y.-C., Rashidi, A., and Rezazadeh Azar, E. (2020). “Automated methods for activity recognition of construction workers and equipment: State-of-the-art review.” *Journal of Construction Engineering and Management*, 146(6), 03120002. DOI: 10.1061/(ASCE)CO.1943-7862.0001843
- P2-13.** Wang, X., and El-Gohary, N. (2023). “Deep learning-based relation extraction and knowledge graph-based representation of construction safety requirements.” *Automation in Construction*, 147, 104696. DOI: 10.1016/j.autcon.2022.104696
- P2-14.** Wu, H., Zhong, B., Li, H., Love, P., Pan, X., and Zhao, N. (2021). “Combining computer vision with semantic reasoning for on-site safety management in construction.” *Journal of Building Engineering*, 42, 103036. DOI: 10.1016/j.jobbe.2021.103036
- P2-15.** Zhang, J., and El-Gohary, N. M. (2015). “Automated information transformation for automated regulatory compliance checking in construction.” *Journal of Computing in Civil Engineering*, 29(4), B4015001. DOI: 10.1061/(ASCE)CP.1943-5487.0000427
- P2-16.** Zhong, B., Wu, H., Ding, L., Love, P. E. D., Li, H., Luo, H., and Jiao, L. (2019). “Mapping computer vision research in construction: Developments, knowledge gaps and implications for research.” *Automation in Construction*, 107, 102919. DOI: 10.1016/j.autcon.2019.102919
- P2-17.** Tricco, A. C., Lillie, E., Zarin, W., O’Brien, K. K., et al. (2018). “PRISMA extension for scoping reviews (PRISMA-ScR): Checklist and explanation.” *Annals of Internal Medicine*, 169(7), 467–473. DOI: 10.7326/M18-0850

Total sources individually cited in manuscript: 42 (19 core + 5 transition + 17 baseline + 1 contextual-only arXiv preprint)

Additional Phase 2 baseline papers tracked in synthesis matrix (not individually cited): 17

Total corpus (peer-reviewed only; excludes the Bui et al. arXiv preprint): 58 papers